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ABERDARE TECHNICAL COLLEGE

LEVEL 5,MODULE 3

CAT2,TERM 3

ELEMENT 2:TEST COMPUTER SOFTWARE FUNCTIONALITY

UNIT OF COMPETENCE: INSTALL COMPUTER SOFTWARE

UNIT CODE: 0619 451 06A

Total Marks: 36 **Time:** 1 Hour 30 Minutes

Name: Martin Ndugire Reg. No: 520 Date: 16-10-2026

Total Marks: 40 **Time:** 1 Hour 30 Minutes

QUESTION 1 (8 marks)

(a) Define the term software testing. **(2 marks)**

(b) Define each of the following terms: **(3 marks)**

(i) Error

(ii) Bug (defect)

(iii) Failure

(c) State any three HTTP methods used in REST and give the purpose of each. **(3 marks)**

QUESTION 2 (8 marks)

(a) Explain the difference between verification and validation, giving one example of each. **(4 marks)**

(b) Using the restaurant analogy (customer, waiter, kitchen), explain what an API is and how it works. **(4 marks)**

QUESTION 3 (8 marks)

(a) An age field on a registration form accepts values from 18 to 60 only. **(4 marks)**

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(i) Use equivalence partitioning to identify the valid and invalid partitions. (2 marks)

(ii) Use boundary value analysis to list the six values you would test. (2 marks)

(b) A school website has the base URL https://school.com/api/. State the HTTP method and a suitable endpoint you would use in Postman to: **(4 marks)**

(i) Retrieve the list of students

(ii) Add a new student

(iii) Update a student's details

(iv) Delete a student record

QUESTION 4 (8 marks)

(a) Compare black box testing and white box testing under the following headings: knowledge of code, skills required, coverage, and what each type finds. **(4 marks)**

(b) A login form accepts passwords of 8 to 12 characters. Analyze how static testing and dynamic testing would each detect a missing check for the 12-character maximum, and explain why the two approaches complement each other. **(4 marks)**

QUESTION 5 (4 marks)

A bank is building a new system that must handle financial transactions with strong built-in security, while a sister company is building a public mobile app backend for simple data retrieval.

Evaluate which of REST or SOAP is better suited to each project. Justify your choices with at least two reasons per project. **(4 marks)**

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(1a) Software testing is the process of checking software to identify errors and ensure it works as expected. (2)

(1b) (i) Error → A mistake made by a person when developing Software

(ii) Bug → A fault in software that may cause in correct results

(iii) Failure → when software does not perform as expected. (3)

(1c) GET → Retrives data

(ii) POST → Adds new data

(iii) DELETE → Remove existing data (3)

(2a) ~~Verification~~ validation checks whether the software meets user needs, while verification checks whether the software is being developed according to specifications. (3)

(2b) Customer – the user requesting a service

(ii) Waiter – The API that carries request and responses

(iii) Kitchen – The server that processes the request.

(iv) Food – The results returned to the customer. (4)

(3a) valid partition 18–60.

> invalid partitions – Below 18 and above 60 (3)

(ii) The six test values are; 17, 18, 19, 59, 60, 61

(3b) (i) Retrive the list of students – Get /students

(ii) Add new student – POST/students

(iii) Update student details – Put/student/{id}

(iv) Delete student record – Delete/students/{id} (4)

(4a)

Black-Box testing	White-box testing
> Test software functionality.	Tests internal code & logic
> No coding knowledge is required.	coding knowledge is required.
> checks input & output.	checks code paths and condition
> Example: Testing a log in form	E.g Testing program branches.

(3)

25

(4b) static testing

> Dynamic testing – Runs log in using passwords.

> Missing check – The system may accept passwords longer than 12 characters.

> why they complement each other: static testing finds problem in the code, while dynamic testing confirms how the software behaves. (3)

(5) (a) Bank handling financial transactions: SOAP

(i) supports structured security standards for sensitive transactions

(ii) supports reliable messaging and transaction-related standards

(b) public mobile app retriving simple data: ~~so~~ REST

(i) It is lightweight and uses less band width. (2)

(ii) It is easy to develop and works well with HTTP and JSON

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